

Project Verification Certificate

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1. Repository Information

Field	Value
Repository name	reactor-yield-hackathon
Git branch	master
Final commit hash	c761ccc714241483dc596a22063fbc9aac586917
Release tag	submission_v1
Tag -> commit (verified match)	c761ccc714241483dc596a22063fbc9aac586917
Commit timestamp	2026-07-26T14:52:26+05:30

2. Environment

Field	Value
Python version	3.11.9
Operating system	Windows 10
Architecture	AMD64
requirements_frozen.txt SHA-256	23b2d580e2fc4820a812dc875b13a0a5f2b321ce3f3206a01a109fe36e6dc7ee
Random seed	42

Package	Version
scikit-learn	1.9.0
numpy	2.4.6
pandas	2.3.3
scipy	1.17.1
catboost	1.2.10

xgboost	3.2.0
lightgbm	4.7.0
optuna	4.9.0
shap	0.51.0
joblib	1.5.3

3. Final Model

Architecture: LinearBlendEnsemble (src/models/ensemble.py)

Feature configuration: core set, 10 features: flow_rate_L_min, concentration_mol_L, inlet_temperature_K, length_m, jacket_temperature_K, avg_temp, residence_proxy, residence_sq, delta_T, severity_index

Base model	Ensemble weight
ExtraTrees	1.397958
CatBoost	0.928030
RandomForest	-1.210939
<i>intercept</i>	-5.294243

Clipping policy: [0.0, 100.0]

Serialized artifact:

artifacts/tuned_pipelines/FINAL_ENSEMBLE_blend_v1.joblib

13,106,831 bytes -- SHA-256: 67d5b7f6237ddc6daf6c8de5a8b8ee7746057b62fd7bd9d718b0b35672099e14

4. Final Validation Protocol

Primary reporting protocol: RepeatedKFold(n_splits=5, n_repeats=10, random_state=42) (50 total folds)

Final ensemble protocol: Leave-one-repeat-out nested cross-validation (10 iterations: train meta-model on 9 of 10 repeats' pooled out-of-fold predictions, evaluate on the 10th, repeated for all 10 choices)

Statistical tests performed: Paired t-test (all major model/tuning/ensemble comparisons), Wilcoxon signed-rank test (non-parametric confirmation of every paired t-test), Bootstrap 95% confidence intervals (5000 resamples, on every paired comparison), Friedman test (13-model omnibus comparison, $\chi^2=530.2$, $p=8.2e-106$), Nemenyi post-hoc test (multiple-comparison-corrected pairwise model comparison), Shapiro-Wilk (residual normality diagnostics), Spearman correlation (heteroscedasticity diagnostics), Kolmogorov-Smirnov test (train/test distribution shift), Population Stability Index (train/test distribution shift, quantile-binned)

5. Final Performance

Model	RMSE	MAE	R2
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Dummy (predict mean)	38.411 +/- 2.600	35.573	-0.0314
Best single model (ExtraTrees (Phase 4 defaults))	16.693	11.772	0.7983
Final ensemble	14.765 +/- 0.634	9.113	0.8511

Improvement over dummy baseline: 61.56%

Improvement over best single model: 11.55%

Statistical significance vs. best single model: paired t-test $p=1.40e-11$, 95% CI=[-2.026, -1.847] (excludes zero: True)

6. Artifact Inventory

Filename	Purpose	Size	SHA-256 (prefix)
submission/TeamName.csv	Final submission CSV	392 B	01d734ea5b404fd6...
.../FINAL_ENSEMBLE_blend_v1.joblib	Final serialized ensemble model	13,106,831 B	67d5b7f6237ddc6d...
...aTrees_SELECTED_FINAL_v1.joblib	ExtraTrees base pipeline (Phase 4 defaults)	5,514,314 B	b57235f543963a75...
...elines/CatBoost_tuned_v1.joblib	CatBoost base pipeline (Phase 5 tuned)	3,197,996 B	2ec8ec5aa5c1a003...
...es/RandomForest_tuned_v1.joblib	RandomForest base pipeline (Phase 5 tuned)	4,396,106 B	e2f866b0006fcfc0...
manifest.json	Machine-readable freeze manifest	4,354 B	a99fb069f6946454...
...ooks/competition_notebook.ipynb	Executed competition notebook	19,151 B	f5ff98d3ce3e58c5...
reports/experiment_registry.md	Full experiment log (EXP-000 to EXP-023)	25,865 B	7f843c5529ef8a41...
reports/feature_registry.md	Feature validation registry	10,451 B	6fb46c94e9f19dd9...
reports/preprocessing_registry.md	Preprocessing decision registry	5,809 B	87cbd0e525825dbe...
reports/final_independent_audit.md	Phase 9 adversarial audit	23,551 B	94475c22956394c3...
reports/technical_defense.md	Phase 8 technical defense	49,976 B	5be059aec1831259...
...ts/optuna_studies/ExtraTrees.db	ExtraTrees Optuna study	229,376 B	24f38a814980c437...
...acts/optuna_studies/CatBoost.db	CatBoost Optuna study	200,704 B	c5f321aeb2a7e90d...
.../optuna_studies/RandomForest.db	RandomForest Optuna study	225,280 B	25567193a33c6ae8...
...tuna_studies/GaussianProcess.db	GaussianProcess Optuna study	139,264 B	486a9706c6df6528...

data/raw/train_dataset.csv	Original training data (150 rows)	5,510 B	b8b499bebf0f0edf...
data/raw/test_dataset.csv	Original test data (50 rows)	1,594 B	2a0b9f57a5c0e8e1...
requirements_frozen.txt	Pinned package versions	1,946 B	23b2d580e2fc4820...
src/inference/submission.py	Submission-writing code (Phase 9 bugfix)	1,327 B	9e63bd00bb1b32c4...
src/inference/validator.py	Validation code (Phase 9 bugfix)	7,552 B	0774bfd0d2c5e67e...
src/models/ensemble.py	LinearBlendEnsemble class definition	1,730 B	bc7dba049e66fed9...

7. Submission Verification

Check	Status	Note
Correct filename	FAIL	Currently 'TeamName.csv' -- placeholder, must be renamed to actual team name before upload
Correct column name	PASS	'overall_yield', verified
Correct row count	PASS	50 rows, verified
Correct ordering	PASS	Matches data/raw/test_dataset.csv row order (pipeline does not reorder)
Utf8 encoding	PASS	Verified via validator.validate_submission_file
Prediction formatting	PASS	Fixed Phase 9: float_format='%.3f' forces exactly 3 decimal places on every row (previously 25/50 rows displayed fewer)
Predictions clipped correctly	PASS	0 values outside [0,100]; n_values_changed_by_clip=0 on the live re-run (internal ensemble clip already correct)
Deterministic inference	PASS	3/3 fresh runs this session produced identical SHA-256: 01d734ea5b404fd60153c7f87050afc5cb886ac9871710
Schema validation passed	PASS	11/11 checks passed on live re-run
Output validation passed	PASS	5/5 prediction checks + 7/7 submission-file checks passed on live re-run (includes the new decimal-place check)
Reproducibility verified	PASS	Re-verified this session, not assumed from a prior phase

8. Validation Checklist

Item	Status
Feature engineering frozen	PASS
Model frozen	PASS
Hyperparameters frozen	PASS
Ensemble frozen	PASS
Reproducibility verified	PASS
Submission validated	PASS
Documentation complete	PASS
Presentation complete	PASS
Technical defense complete	PASS
Final audit complete	PASS
Critical defects discovered during development resolved	PASS
Administrative action pending (team name placeholder)	FAIL

9. Known Limitations

1. Feature selection (which of 24 candidates to engineer/keep) was performed using statistics computed on the full 150-row training set, not nested inside cross-validation.

Evidence: reports/technical_defense.md Part A (Phase 2); reports/final_independent_audit.md Section 2

Expected impact: Likely small optimistic bias in reported CV performance; magnitude never directly quantified via a nested-feature-selection re-run.

Mitigation: Disclosed explicitly in 2 independent reports rather than hidden; not corrected in the shipped model.

2. Approximately 300-500 statistical hypothesis tests were run across the exploratory phases (Phase 1-2) without a formal project-wide multiple-comparisons correction (e.g. Bonferroni or FDR).

Evidence: reports/final_independent_audit.md Section 2

Expected impact: At $\alpha=0.05$, roughly 15-25 spuriously 'significant' exploratory findings would be expected by chance alone. Headline claims (e.g. the final ensemble improvement, $p<0.000001$) are large enough to very likely survive any reasonable correction; borderline findings (e.g. severity_index at $p=0.05$ in one specification) are less certain.

Mitigation: Informal multi-test promotion bar used (features required to pass multiple independent tests, not one); no formal correction applied.

3. The final reported RMSE (14.76) is the best result from an iterative search across 23 logged experiments (EXP-000 to EXP-023) -- a 'winner's curse' / researcher-degrees-of-freedom effect distinct from any single experiment's own validity.

Evidence: reports/final_independent_audit.md Section 2; reports/experiment_registry.md

Expected impact: Small optimistic lean on the specific reported number; the underlying comparison's effect size ($p < 0.000001$) is large enough that this is unlikely to change the qualitative conclusion.

Mitigation: Named explicitly for the first time in the Phase 9 audit; not correctable post-hoc without independent data.

4. The final ensemble's coefficients (including a negative weight on RandomForest) are validated in-distribution only -- every robustness check (10-way repeat-swap, leave-one-repeat-out nested CV) used the same 150 training rows used throughout the entire project.

Evidence: reports/technical_defense.md Part A (Phase 6); reports/ensemble_evaluation_report.md Section 4-5

Expected impact: Unknown behavior on data meaningfully different from the training distribution; this is the single largest open risk in the project.

Mitigation: Robustness within the available data was verified as rigorously as this dataset allows (10 independent refits, full leave-one-repeat-out nested validation); no out-of-distribution test exists.

5. The frozen environment (requirements_frozen.txt) was validated only in the single virtual environment used throughout this project -- never independently reproduced in a second, freshly created environment.

Evidence: reports/competition_readiness_report.md Section 2; reports/final_independent_audit.md Section 4

Expected impact: A judge running outside the provided requirements file (e.g. their own existing environment with different numpy/scikit-learn versions) could encounter API incompatibilities.

Mitigation: Exact-pin requirements_frozen.txt provided; residual risk depends on whether it is actually used.

6. 42% of test-set predictions (21/50) are exactly 0.0 (clipped), versus a 24.7% zero-yield rate in training. Cannot be fully resolved without the true test labels.

Evidence: reports/submission_validation_report.md; reports/confidence_audit.md row 24; reports/presentation_outline.md backup slide B1

Expected impact: Test rows predicted zero have a mean avg_temp of 467.6K, closely matching the training zero-yield-group mean of 469.1K -- evidence favoring 'model correctly recognizes the same physical regime' over 'arbitrary over-conservatism', but not proof.

Mitigation: Investigated with real data rather than left as an open question; documented as a prepared backup slide.

7. The exact physical mechanism behind the zero-yield regime (over-reaction via k2-dominance vs. a possible simulator/solver artifact) was never independently verified against true reaction kinetics.

Evidence: reports/phase1_eda_findings.md Section 6 challenge table; reports/final_independent_audit.md Section 1

Expected impact: Does not affect model performance (the regime's existence is statistically well-supported); affects only the confidence with which a physical mechanism can be claimed in presentation.

Mitigation: Presentation materials instructed to use 'consistent with' rather than 'confirms' language for this specific claim.

10. Risk Register

Risk	Likelihood	Impact	Residual Risk
TeamName.csv placeholder not renamed to actual team name before upload	Medium	High (could void/misdirect the submission)	Low if acted on before upload; High if forgotten
Judge's environment differs from requirements_frozen.txt	Medium	Medium (import/runtime failure, not silent wrong answers)	Low-Medium, untested on a second machine
Feature-selection performed on full dataset, not nested in CV	High (definitely true)	Low-Medium (unquantified optimistic bias)	Low-Medium, unquantified
Multiple-comparisons exposure across exploratory phases	High (definitely true)	Low for headline claims, Medium for borderline ones	Low for the shipped model's core claims
Ensemble coefficients validated in-distribution only	High (structurally true)	Unknown (untested out-of-distribution)	Medium-High for any future deployment beyond this exact dataset
Undiscovered bugs beyond the 2 found via adversarial testing (Phase 7 validator crash, Phase 9 decimal-format defect)	Unknown	Unknown	Cannot be reduced to zero; no further mitigation identified

11. Certification Statement

This project has completed domain analysis, feature engineering, preprocessing validation, model benchmarking, hyperparameter optimization, ensemble evaluation, reproducibility verification, submission validation, and independent adversarial audit. All critical defects discovered during development -- including one found during this certificate's own preparation (a submission decimal-formatting compliance bug) -- have been resolved and re-verified live, not merely assumed fixed. The submission artifact identified by SHA-256 hash 01d734ea5b404fd60153c7f87050afc5cb886ac987171da4e18fd978a04b1390 is the exact artifact intended for competition submission, produced by commit c761ccc714241483dc596a22063fbc9aac586917 (tag: submission_v1). Unless a competition rule changes or a new critical defect is discovered, no further technical modifications are recommended.

12. Final Recommendation

READY AFTER ADMINISTRATIVE ACTION

Required administrative action(s):

1. Rename submission/TeamName.csv (and src/inference/config.py:TEAM_NAME) to the actual competition team name before upload. This is the only remaining action item; it requires no code logic changes and does not

require re-running inference validation (the CSV content is identical regardless of filename).

Rendered from project_verification_certificate.json (single source of truth for all 3 certificate formats). Repository: reactor-yield-hackathon | Commit: c761ccc71424 | Tag: submission_v1